

WORKPACK 5

Merge and Append

Level 3 Data Technician · GreenWheels Urban Mobility · Self-Diagnostic Pack

Get Ready to Smash This

A head-start check — this is not a test

Below are eight things this pack covers. Tick the ones you can already do.

If you can tick all eight: jump to the Mini-Mission on page 9 and prove it. Then move on to Pack 6.

- ☐ I know the difference between Append Queries and Merge Queries in Power Query.
- ☐ I can use Append Queries to stack two datasets with the same columns on top of each other.
- ☐ I can use Merge Queries (Left Outer Join) to add columns from a lookup table to a main dataset.
- ☐ I know what a join key is and how to select the correct column(s) for a join.
- ☐ I can explain what a Left Outer Join, Inner Join, and Left Anti Join each return.
- ☐ I know what a composite key is and how to set up a merge using two columns.
- ☐ I can expand a merged column in Power Query to show the columns I need.
- ☐ I can use a Left Anti Join to find records in one table with no match in another.

Marcus says: "Priya told me to merge the zone lookup table. I copied it into a new sheet and did a VLOOKUP. Is that the same thing?" Let's see why Power Query merge is more reliable.

What This Pack Is About

Real-world data is almost never stored in a single table. GreenWheels has hire records in one file and zone management details in another. To build a complete picture, you need to combine them. Power Query has two operations for this: Append (stack rows) and Merge (add columns from another table).

You will use Pack5_GW_data.xlsx, which contains three sheets: '2024 Data' (40 clean records), 'Q1 2025 Data' (10 new records from the next year), and 'Zone Lookup' (5 rows of zone manager information). The exercises build on each other — do them in order.

Append vs Merge — the key difference

NOTE

Append — adds more rows. Stack two tables that have the same columns. Like adding new rows to the bottom of a spreadsheet.

Merge — adds more columns. Match rows from two tables using a shared key. Like a VLOOKUP that brings in whole columns, not just one value.

The Skills

Open Pack5_GW_data.xlsx. Load all three sheets into Power Query as separate queries before you start Chunk 1. Go to Data → Get Data → From File → From Workbook. In the Navigator, tick all three sheets, then click Transform Data.

1

Append Queries

Stacking two years of data

SCENARIO

"Q1 2025 data has just arrived," says Priya.

"I need one table with all of 2024 and Q1 2025 together so I can compare quarter by quarter. Can you stack them?"

Append Queries places all the rows from one query underneath the rows of another query. Both queries must have the same column names for the columns to line up. Where a column exists in one query but not the other, Power Query fills with null.

WORD TO KNOW

Append Queries

Home tab → Append Queries. Choose two (or more) queries to stack. The columns must have matching names. The result is one combined query with all rows from both sources.

WORD TO KNOW

Primary query / Secondary query

When appending, the Primary query's rows come first. The Secondary query's rows are added below them. You choose which is which in the dialog.

To append: with the '2024 Data' query selected, go to Home → Append Queries → Append Queries as New. Choose '2024 Data' as the first table and 'Q1 2025 Data' as the second. A new query is created containing all rows from both.

TRY IT

In Power Query, select the '2024 Data' query. Go to Home → Append Queries → Append Queries as New. Set the first table to '2024 Data' and the second to 'Q1 2025 Data'. Click OK.

A new query called 'Append1' appears. Rename it 'All Hire Data'. Count the rows — you should have 50 records (40 from 2024, 10 from Q1 2025).

Sort by Hire Record ID ascending. Confirm that HC-043 to HC-052 appear at the bottom, with 'Q1 2025' in the Period column.

CHECK YOURSELF

What would happen if the Q1 2025 sheet had a column called 'Average Duration' instead of 'Avg Hire Duration (Mins)'?

Answer: Power Query would not match them as the same column. The appended table would have both as separate columns — 'Avg Hire Duration (Mins)' filled with nulls for Q1 2025 rows,

and 'Average Duration' filled with nulls for all 2024 rows. Column names must match exactly for Append to work correctly.

2

Merge Queries — Left Outer Join

Adding zone manager details to the hire data

SCENARIO

"I need to know which manager is responsible for each hire record," says Brian.

"The Zone Lookup table has the zone manager names. Can you bring that into the main dataset using the Zone column as the link?"

Merge Queries matches rows from two tables using a shared key column. The Left Outer Join keeps all rows from the left table (your main data) and adds matching columns from the right table (the lookup). If a row in the left table has no match in the right, the added columns are null.

WORD TO KNOW

Merge Queries

Home tab → Merge Queries. Select the matching column(s) in each table, choose the join type, and click OK. The right table is added as a new column of type 'Table'. You then expand it to choose which columns to bring in.

WORD TO KNOW

Join key

The column (or columns) used to match rows between two tables. In this merge, the join key is Zone — every row in the hire data is matched to the row in the Zone Lookup where Zone equals the same value.

To merge: with the '2024 Data' query selected, go to Home → Merge Queries → Merge Queries as New. Set the left table to '2024 Data', left column to Zone. Set the right table to 'Zone Lookup', right column to Zone. Choose Left Outer as the join type. Click OK.

After clicking OK, a new column called 'Zone Lookup' appears on the right of the table. It contains a Table object for each row. Click the expand icon (two arrows) at the top of that column and choose which columns to bring in — for example, Zone Manager and Quarterly Target.

TRY IT

Perform the Left Outer merge as described above. Expand the Zone Lookup column — bring in 'Zone Manager' and 'Quarterly Target (Hires)'. Deselect 'Use original column name as prefix'. Click OK.

Check HC-001: Zone = Central. Zone Manager should now show 'Priya Sharma'. Quarterly Target should show 500.

How many rows are in the merged query? (Should still be 40 — merge adds columns, not rows.)

CHECK YOURSELF

After a Left Outer merge, why might some rows have null in the Zone Manager column?

Answer: A Left Outer Join returns null in the added columns when a row in the left table has no matching value in the right table. If a hire record's Zone value does not appear in the Zone Lookup table, there is no manager to bring in — so those cells show null.

3**Understanding Join Types***What each one keeps and what it drops***SCENARIO***"I keep hearing about left joins, right joins, inner joins," says Marcus.**"I just pick Left Outer every time because it feels safe. But I don't really know what I'm choosing." Let's fix that.*

Every merge in Power Query requires a join type. The join type decides which rows from each table appear in the result. Here are all six types, with what they mean in GreenWheels terms.

Join Type	What it keeps	GreenWheels example result
Inner Join	Only rows where the key matches in both tables	40 records — all 2024 records matched to a zone
Left Outer	All rows from the left table, plus matching rows from the right	40 records — nulls if any zone had no match in lookup
Right Outer	All rows from the right table, plus matching rows from the left	Up to 5 rows — one per zone in the lookup
Full Outer	All rows from both tables, nulls where there is no match	40 + any unmatched zones
Left Anti	Rows in the left table that have NO match in the right table	Any records with a zone not in the lookup
Right Anti	Rows in the right table that have NO match in the left table	Any zones in the lookup with no 2024 records

The most common join in data work is Left Outer — it keeps your main data intact and adds matched lookup values. Inner Join is useful when you only want rows that successfully matched. Anti joins are useful for finding gaps and mismatches.

TRY IT

Perform a new merge: '2024 Data' left, 'Zone Lookup' right, join on Zone. This time choose Inner Join. How many rows do you get? (You should still get 40 — because every zone in the hire data does exist in the lookup.)

Now try Left Anti Join. How many rows? What does this tell you about the data quality of the Zone column?

Try Right Anti Join. How many rows? What would it mean if you got any rows here?

CHECK YOURSELF

You get 0 rows from a Left Anti Join between '2024 Data' and 'Zone Lookup' on Zone. What does that tell you?

Answer: Every zone value in the hire data matches a zone in the lookup. There are no records with an unrecognised zone. The Zone column is consistent and complete — a useful data quality check.

4

Composite Keys*Merging on two columns at once***SCENARIO**

"I want to check whether any station appears in the 2024 data without any record in Q1 2025," says Priya.

"I need to match on Station Name and Zone together — not just one column."

A composite key is a join that uses two or more columns together as the match. If you join on Station Name alone and two different zones share the same station name, every row would match incorrectly. Adding Zone as a second key column ensures you only match rows where both the name and the zone are identical.

WORD TO KNOW**Composite key**

A join where two or more columns must match at the same time. In Power Query, you select the first key column in each table, then hold Ctrl and click the second key column to add it. Power Query shows both columns highlighted.

To set up a composite key merge: go to Home → Merge Queries as New. Choose the left table, click the first key column (Station Name). Hold Ctrl and click the second key column (Zone). Do the same for the right table. Power Query labels each selected column with a small number (1, 2) to show the matching order.

TRY IT

Create a new merge: left table = 'Q1 2025 Data', right table = '2024 Data'. Join type = Right Anti. For the join key, select Station Name only. How many rows do you get?

Now add Zone as a second key column (Ctrl + click). Run the merge again with Right Anti. Does the row count change? What does this tell you about the data?

A Right Anti Join here returns stations that are in the 2024 data but have no matching record in Q1 2025. Are any stations missing from Q1 2025?

CHECK YOURSELF

Why might a merge on Station Name alone give wrong results if two stations happened to have the same name in different zones?

Answer: A match on station name alone would pair those two stations together, treating them as the same location. The join would bring in data from the wrong station. Adding Zone as a second key ensures the match requires both the name AND the zone to be identical — preventing false matches.

5**Expand, Select, and Rename***Finishing the merged query***SCENARIO**

"When I expanded the merge column, I got all five columns from the Zone Lookup, including ones I don't need," says Brian.

"And the column names now say Zone Lookup.Zone Manager. How do I tidy this up?"

After expanding a merged column, Power Query shows every column from the right table. You can choose only the columns you need during the expand step. If you forgot to deselect 'Use original column name as prefix', you get prefixed names — for example 'Zone Lookup.Zone Manager'. You fix this by renaming the column.

WORD TO KNOW**Expand step**

After a merge, the new column contains a Table object. Click the expand icon (two outward arrows) in the column header to choose which columns to bring in. Deselect 'Use original column name as prefix' to get clean column names.

WORD TO KNOW**Remove Columns**

Right-click a column header → Remove (or select multiple columns with Ctrl, then right-click → Remove Columns). Removes the columns from the query without affecting the source data.

TRY IT

Go back to the merge query you created in Chunk 2. Click the expand icon on the Zone Lookup column. This time, select only 'Zone Manager' and 'Quarterly Target (Hires)'. Make sure 'Use original column name as prefix' is unchecked. Click OK.

Rename the columns if they still have prefixes: double-click the column header and type the clean name.

Remove any columns you do not need for the analysis. A good final table for this exercise: Hire Record ID, Station Name, Zone, Period, Total Hires, Total Revenue, Zone Manager, Quarterly Target.

CHECK YOURSELF

What is the quickest way to remove several columns at once in Power Query?

Answer: Hold Ctrl and click each column header to select them all. Then right-click any selected header → Remove Columns. All selected columns are removed in one step, recorded as a single step in Applied Steps.

Spot the Mistake

Can you see what went wrong?

Marcus merges '2024 Data' with 'Zone Lookup' using Zone as the join key. He chooses Inner Join. He loads the result and shows Priya — the table has 40 rows.

Next day, the '2024 Data' is updated. Someone has added five records for a new zone called 'South-East' which does not yet appear in the Zone Lookup. Marcus refreshes. The loaded table now shows only 40 rows — the five new records are missing.

WATCH OUT

Inner Join silently drops any rows that have no match in the lookup. The five new 'South-East' records have no match in Zone Lookup, so they disappear. Marcus does not notice because the table refreshed without errors. Left Outer Join would keep all rows and show null for the missing zone details — making the gap visible.

CHECK YOURSELF

What join type would Marcus need to use to ensure all hire records are kept, even if their zone is not yet in the lookup?

Answer: Left Outer Join. It keeps all rows from the left table (the hire data) regardless of whether a match is found on the right (the zone lookup). Missing matches show null rather than being silently removed.

Mini-Mission

Two minutes — prove it

Using Pack5_GW_data.xlsx: (1) Append the 2024 data and Q1 2025 data into one 50-row query. (2) Merge the 2024 data with the Zone Lookup using a Left Outer Join on Zone. Expand to show Zone Manager and Quarterly Target. (3) Run a Left Anti Join to confirm zero unmatched zones. Close and load all three queries.

You Can Now Answer

These are the kinds of questions that come up in Scenario Demonstration 2

- ☐ What is the difference between Append Queries and Merge Queries?
- ☐ When would you use Append Queries? Give an example.
- ☐ What is a join key, and why does it have to match in both tables?
- ☐ What does a Left Outer Join keep, and what does it set to null?
- ☐ What is a Left Anti Join useful for?
- ☐ What is a composite key, and how do you set one up in Power Query?
- ☐ How do you select which columns to bring in when expanding a merged column?

If You Got Stuck

No problem — here is what to practise next

Merge Queries clicked for me when I stopped thinking about SQL joins and just asked: am I adding rows (Append) or adding columns (Merge)? The join type only matters for the merge.

— Priya, **GreenWheels Data Lead**

If join types still feel confusing: look at the Join Types table in Chunk 3 and create each one using the GreenWheels data. Count the rows each time. Seeing the row counts change is the fastest way to understand what each join does.

If composite keys caused problems: remember to hold Ctrl when clicking the second key column in each table. The small number badges (1 and 2) confirm both columns are selected.

Pack 6 is the last pack: Spot Simple Patterns. You will use the clean, merged data to make plain-English observations about what the GreenWheels numbers actually show.